

Harrison Neill

Adelaide, SA ♦ 0437 810 116 ♦ harryneill930@gmail.com ♦ [LinkedIn](#) ♦ [Project Portfolio](#)

CAREER OBJECTIVE

Final year mechanical engineering student and Structural Subsystem Lead at Adelaide University Solar Racing, experienced in building and racing solar powered vehicles for the Bridgestone World Solar Challenge. Practical experience in CAD, FEA, CFD, and carbon fibre composite fabrication. Seeking graduate engineering roles commencing in 2027 where I can contribute to product design and advanced manufacturing.

EDUCATION

Bachelor of Engineering (Honours) (Mechanical): 2022 – 2026 (expected)

Adelaide University:

Key Courses: Engineering Research Project Part A (HD), Advanced Manufacturing Systems (D), Advanced Vibrations (D)

KEY SKILLS

- **Engineering Software:** Autodesk Inventor, Autodesk Alias, AutoCAD, Design Modeler, Spaceclaim
- **Simulation:** ANSYS Mechanical (FEA) and ANSYS Fluent/CFX (CFD)
- **Programming:** MATLAB, Python
- **Manufacturing:** Experienced in carbon fibre composite layup, vacuum bagging, and resin infusion
- **Certifications:** White card (Construction Induction)

PROJECTS

Lumen IV - Structural Subsystem Lead, AU Solar Racing, Final Year Research Project, April 2026 – Present

- Leading the design of the aeroshell of Lumen IV, utilising Autodesk Alias and CFD to optimise the aerodynamic performance of the vehicle, reducing the drag coefficient by 20% and weight by 26% (relative to Lumen III) at the design stage
- Managing 12 students across 5 teams (canopy, roll hoop, clamshells, material testing, occupant cell), coordinating integration with the suspension, steering, battery pack and wiring harness subsystems

Streamlined Drone – Personal Project, June 2026 – present

- Designing a battery powered FPV quadcopter drone targeting the straight-line velocity world record of 657 km/h
- Developed parametric performance model in Python (thrust balance, pitch) identifying a predicted 665 km/h configuration and generated the body geometry using a coordinate generator in MATLAB
- Designed structural frame and electrical powertrain in Autodesk Inventor, utilising a sled system so the entire power train can be removed from the body and worked on separately

Lumen III - Team Member, AU Solar Racing, March 2024 – August 2025

- Designed and built retractable fairing pivot mechanism for Lumen III, ensuring vehicle dynamics met the required 7 meter turning circle
- Applied FEA in ANSYS to ensure designs met regulations and achieved a safety factor of 3
- Competed in the Bridgestone World Solar Challenge with the fairing system performing reliably during the race

EXPERIENCE

Installer, *CPR Safe Ind*, November 2025 – Present

- Interpreted engineering drawings to accurately install safety barriers in warehouses and factories
- Collaborated with clients to solve installation problems and achieve compliant outcomes, performing quality checks to ensure installations met client needs

Composites Engineering Intern, AU Solar Racing (hosted at Etamax Engineering), Jan 2025 - July 2025

- Fabricated the carbon fibre body, ribs, torsion box, and clam shells of Lumen III at Etamax Engineering's commercial workshop, working with their engineers
- Optimised component strength to weight ratio, calculated resin requirements and examined part quality
- Worked safely with hazardous materials such as industrial grade adhesives, carbon fibre, fibre glass, epoxy resin, Utilising correct PPE and safe operating procedure